HIMANSHU GAUR

SR. AI ENGINEER

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PROFESSIONAL SUMMARY

AI / Machine Learning Engineer with **4 years** of professional experience building and deploying **machine learning, computer vision, NLP, and edge AI solutions**. Currently serving as **Sr. AI Engineer at PayMagen Informatics and Automation**, working on real-time edge vision and the **Interceptor Drone Tracking Algorithm** using Hailo-8, Hailo-8L, YOLOv8, HailoRT, OpenCV and multi-object tracking. Previous experience at BYTES SOFTWARE spans predictive analytics, recommendation systems, anomaly detection, ML pipelines, NLP and cloud-based solutions. Strong in Python, PyTorch, TensorFlow, Scikit-learn, SQL, FastAPI, MLOps and model optimization.

PROFESSIONAL EXPERIENCE

Sr. AI Engineer | PayMagen Informatics and Automation

01 June 2026 – Present | Delhi, India

- Delivered the Interceptor Drone Tracking Algorithm project as a real-time edge AI computer vision system using Hailo-8 and Hailo-8L for accelerated, low-latency inference.
- Integrated YOLOv8, HailoRT, OpenCV and GStreamer with multi-object tracking to build a real-time perception pipeline; project testing reported 30+ FPS and sub-50 ms end-to-end perception latency.
- Designed modular AI components for detection, tracking, identity, threat assessment, decision logic, event logging and alerting, supporting deployment-oriented edge vision workflows.
- Optimized inference for edge hardware through model optimization, hardware acceleration, asynchronous processing and custom post-processing techniques.
- Investigated and resolved inference, pipeline integration and runtime issues through systematic testing, profiling and troubleshooting.

Machine Learning Engineer | BYTES SOFTWARE

27 February 2023 – 29 May 2026 | Pune, India

- Engineered and deployed machine learning models for predictive analytics, recommendation systems and anomaly detection, with reported improvements of 15% in prediction accuracy and 10% reduction in false positives.
- Performed data preprocessing, feature engineering, dimensionality reduction, cross-validation and hyperparameter tuning, reducing reported training time by 20% and improving F1-score by 5%.
- Built and maintained ML pipelines for data ingestion, model training, validation and deployment using MLflow and Kubeflow; reported 40% faster deployment turnaround and CI/CD support for 100K+ daily records.
- Evaluated models using AUC, precision, recall and F1-score, with reported performance reaching 95% AUC.
- Collaborated with product, engineering and business stakeholders to translate requirements into practical AI solutions and resolve production issues, contributing to a reported 30% reduction in downtime.

TECHNICAL SKILLS

Programming: Python, SQL

Machine Learning: Scikit-learn, XGBoost, LightGBM, Random Forest, Gradient Boosting, SVM, Logistic Regression, feature engineering, hyperparameter tuning, model evaluation

Deep Learning: PyTorch, TensorFlow, Keras, CNNs, RNNs, Neural Networks

Computer Vision & Edge AI: YOLOv8, OpenCV, Hailo-8, Hailo-8L, HailoRT, GStreamer, multi-object tracking, face recognition, real-time inference, edge deployment

NLP & GenAI: SpaCy, NLTK, Gensim, OCR, NER, Transformers, RAG, LangChain, text preprocessing, embeddings, TF-IDF

MLOps & APIs: MLflow, Kubeflow, MLOps, CI/CD, FastAPI, Flask, Git, Docker, Kubernetes

Cloud & Data: GCP BigQuery, Cloud Functions, AWS Lambda, MySQL, MongoDB, SQL Server

Analytics: Power BI, Matplotlib, Seaborn

SELECTED PROJECTS

Interceptor Drone Tracking Algorithm - Edge AI | PayMagen Informatics and Automation

- Real-time edge vision system centered on Hailo-8 / Hailo-8L and Raspberry Pi-class hardware for low-latency AI inference.
- Integrated YOLOv8, HailoRT, GStreamer, OpenCV and multi-object tracking for real-time detection and tracking workflows.
- Developed modular components covering detection, tracking, identity, threat assessment, decision logic, event logging and alerts.

Allstate Insurance Claims Severity Prediction | BYTES SOFTWARE

- Developed machine learning models for insurance claim severity prediction to support risk assessment and resource allocation, with a reported 5% reduction in claims processing costs.
- Compared Random Forest, Gradient Boosting and XGBoost using k-fold cross-validation and feature engineering.
- Generated actionable insights associated with a reported 10% improvement in claims processing efficiency.

Loan Eligibility Prediction using Machine Learning on GCP | BYTES SOFTWARE

- Developed a predictive model for loan eligibility using Google Cloud Platform; performed preprocessing and feature engineering with BigQuery and Apache Beam.
- Evaluated Logistic Regression, Decision Trees and Gradient Boosting with hyperparameter tuning; reported 90% accuracy and deployed using Cloud Functions.

Resume Parsing and Information Extraction - NLP/OCR | BYTES SOFTWARE

- Built an information-extraction pipeline using Python, OCR and SpaCy with text preprocessing, named-entity recognition and section extraction.
- Reported 95% information-extraction accuracy.

Sogeti UK Dashboarding Solution - Power BI | BYTES SOFTWARE

- Developed interactive Power BI dashboards using SQL Server; created and modified stored procedures for complex datasets.
- Automated email notifications and collaborated with clients on custom reporting requirements.

EDUCATION

Master of Business Administration (MBA) | ICCMRT college, Lucknow, Uttar Pradesh | AKTU | 2020 | 67 %

Bachelor of Commerce (B.Com) | University of Jammu, Jammu, India | 2017 | 52%+

ADDITIONAL INFORMATION

Languages: English (Fluent), Hindi (Native)

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